

SHETH L.U.J. AND SIR M.V. COLLEGE

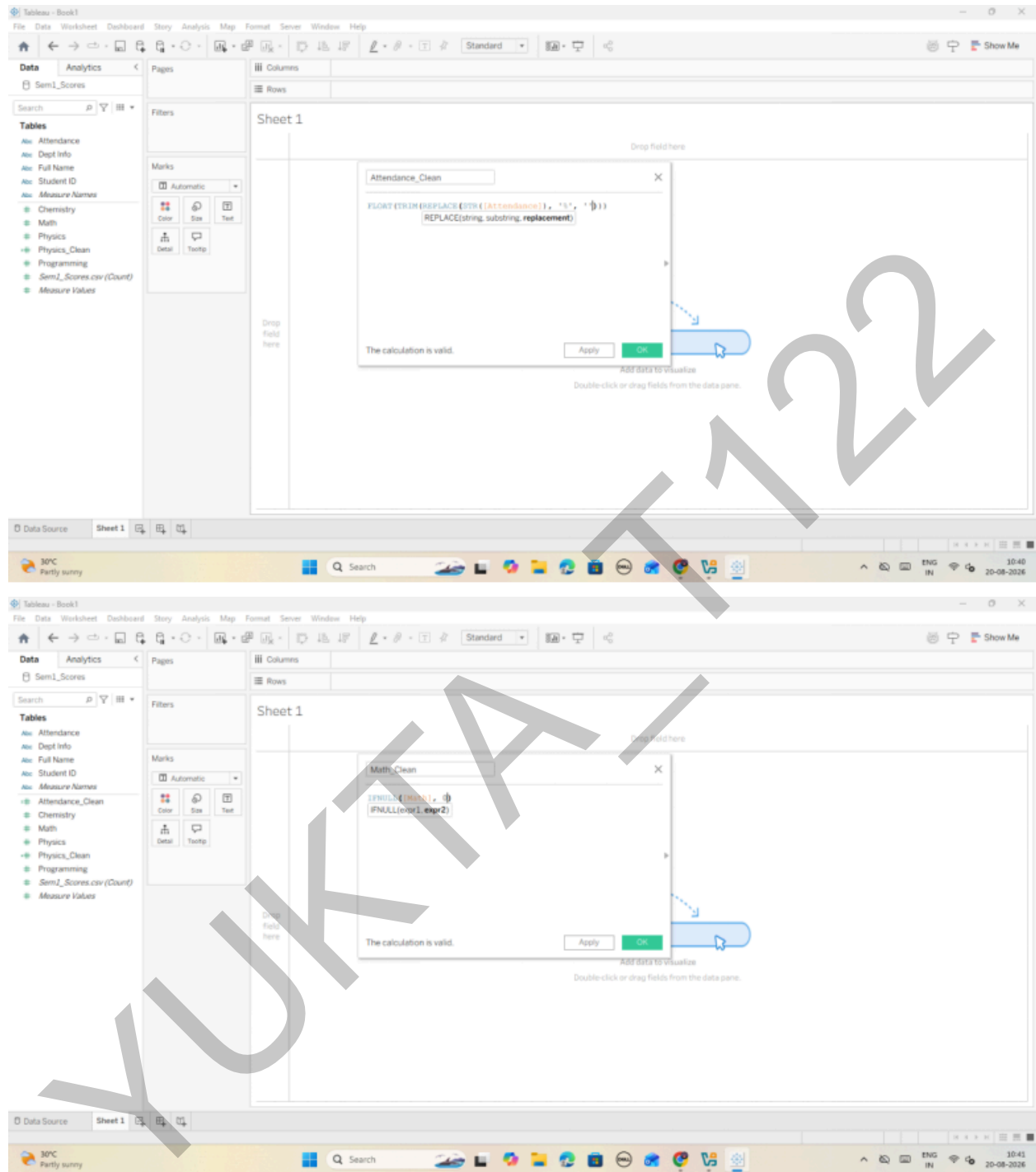
AIM:Cleaning and Structuring Data for Analysis

- Identify and handle missing or inconsistent data
- Convert wide data into tall format

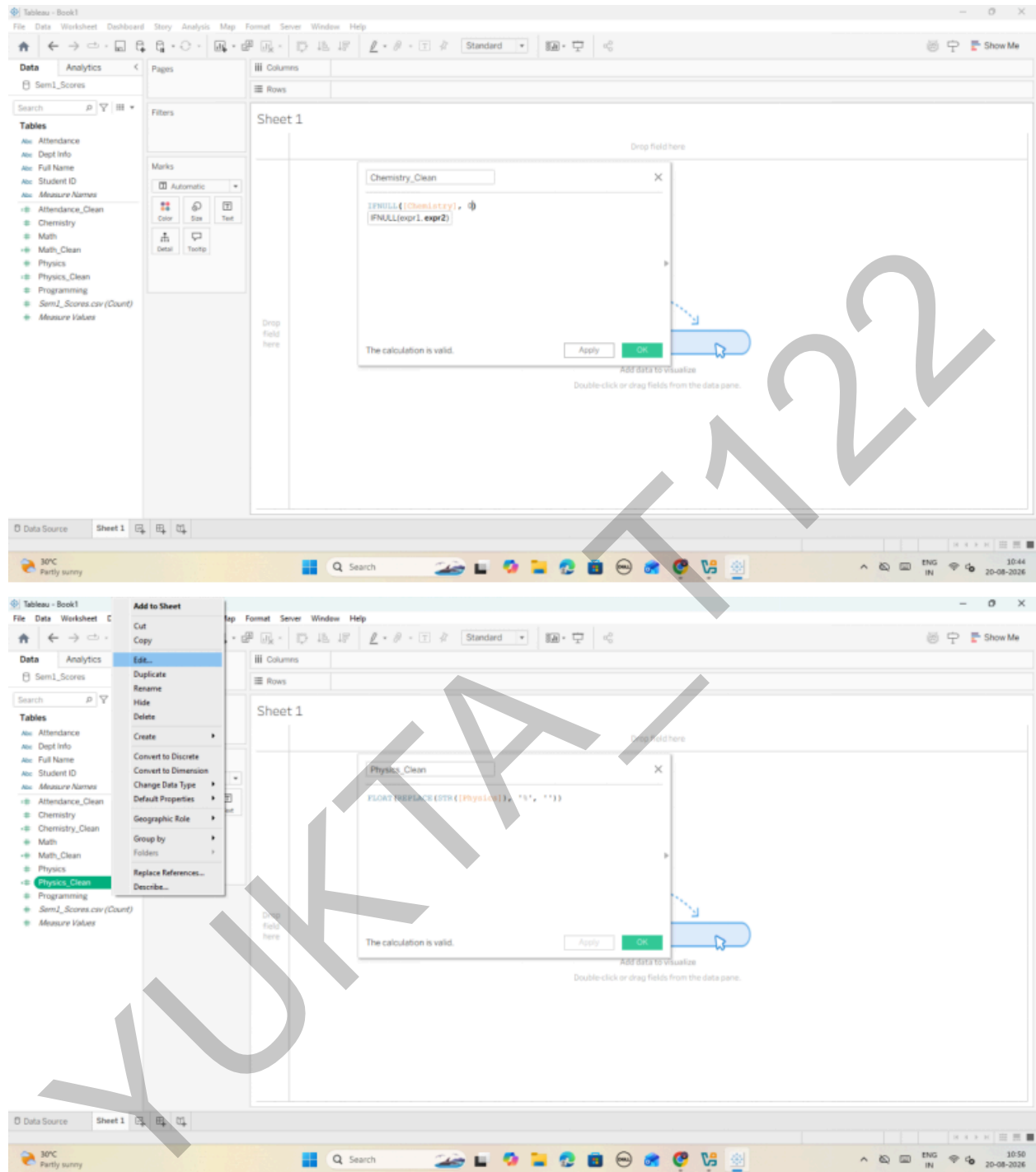
The top screenshot shows the Tableau interface with the 'Sem1_Scores.csv' data source loaded. The data is displayed in a table view with columns for Student ID, Full Name, Dept Info, Math, Physics, Chemistry, Programming, and Attendance. The bottom screenshot shows the 'Physics_Clean' calculated field dialog box, where the formula `ROUND(REPLACE(STR(IFNULL([Physics]), ' '), ' ', ''))` is entered to clean the Physics score field by removing spaces. The dialog box also shows the 'Apply and Close' button.

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S301	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S302	priya VERMA	ECE-02	null	0.700000	80	85	90 %
S303	Rohan Gupta	CSE-01	78	0.520000	null	88	78 %
S304	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S305	aditya Rao	MECH-04	45	null	50	60	65 %
S306	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S307	Rahul Nair	ECE-02	null	0.700000	75	80	72 %

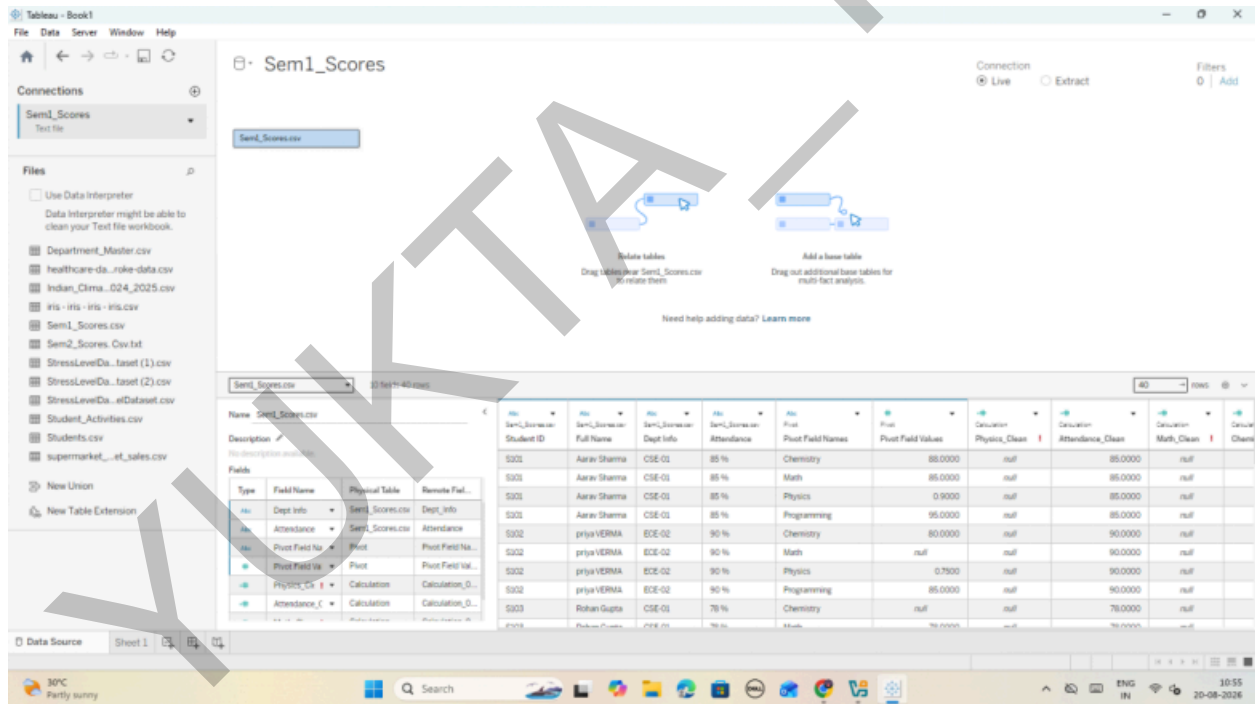
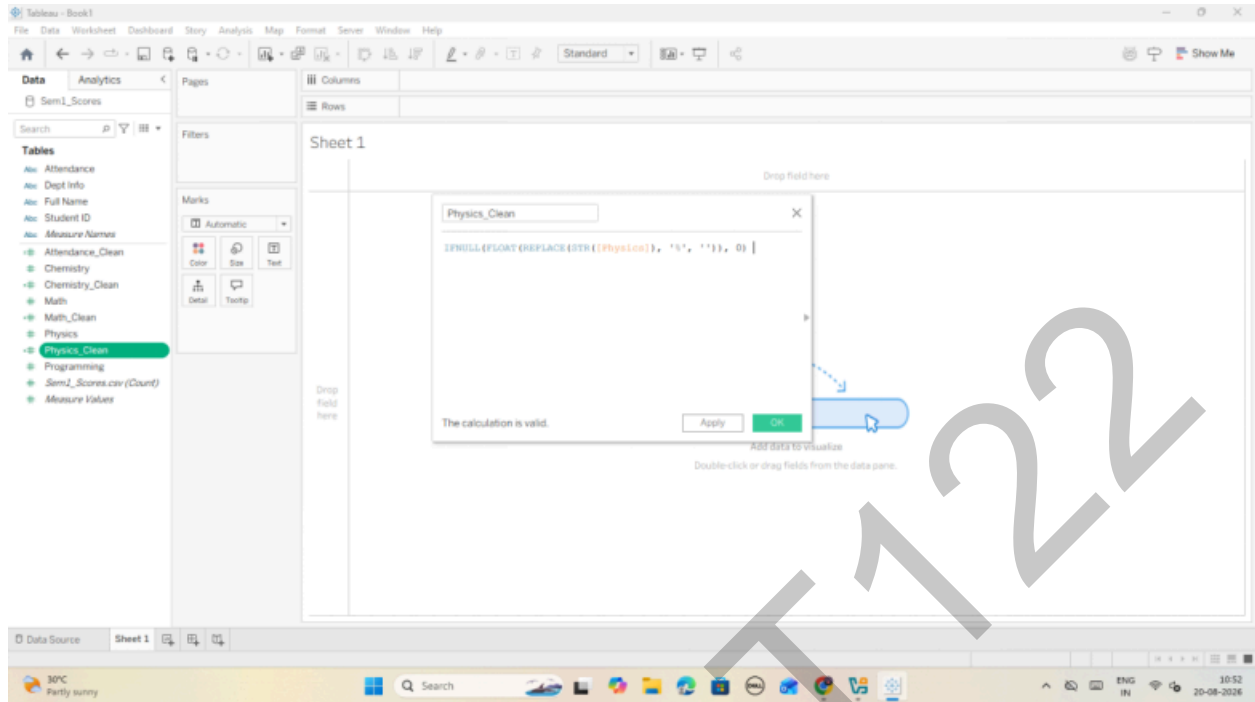
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Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
healthcare-da_roke-data.csv
Indian_Clima_024_2025.csv
iris - iris - iris.csv
Sem1_Scores.csv
Sem2_Scores.Csv.txt
StressLevelDa_taset (1).csv
StressLevelDa_taset (2).csv
StressLevelDa_etDataset.csv
Student_Activities.csv
Students.csv
supermarket_et_sales.csv

New Union
New Table Extension

Sem1_Scores.csv

Description

Fields

Type	Field Name	Physical Table	Remote Field
Text	Attendance	Sem1_Scores.csv	Attendance
Text	Subject	Pivot	Pivot Field Name
Text	Score	Pivot	Pivot Field Value
Text	Physics_Ck	Calculation	Calculation_D
Text	Attendance_C	Calculation	Calculation_D
Text	Math_Ck	Calculation	Calculation_D

10 fields 40 rows

Student ID	Full Name	Dept Info	Attendance	Subject	Pivot Field Value	Physics_Clean	Attendance_Clean	Math_Clean	Chems
S001	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	null	85.0000	null	
S001	Aarav Sharma	CSE-01	85 %	Math	86.0000	null	86.0000	null	
S001	Aarav Sharma	CSE-01	85 %	Physics	0.9000	null	85.0000	null	
S001	Aarav Sharma	CSE-01	85 %	Programming	95.0000	null	85.0000	null	
S002	priya VERMA	ECE-02	90 %	Chemistry	80.0000	null	90.0000	null	
S002	priya VERMA	ECE-02	90 %	Math	0.7500	null	90.0000	null	
S002	priya VERMA	ECE-02	90 %	Physics	0.7500	null	90.0000	null	
S002	priya VERMA	ECE-02	90 %	Programming	85.0000	null	90.0000	null	
S003	Rohan Gupta	CSE-01	78 %	Chemistry	78.0000	null	78.0000	null	
S003	Rohan Gupta	CSE-01	78 %	Math	78.0000	null	78.0000	null	

Need help adding data? Learn more

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
healthcare-da_roke-data.csv
Indian_Clima_024_2025.csv
iris - iris - iris.csv
Sem1_Scores.csv
Sem2_Scores.Csv.txt
StressLevelDa_taset (1).csv
StressLevelDa_taset (2).csv
StressLevelDa_etDataset.csv
Student_Activities.csv
Students.csv
supermarket_et_sales.csv

New Union
New Table Extension

Sem1_Scores.csv

Description

Fields

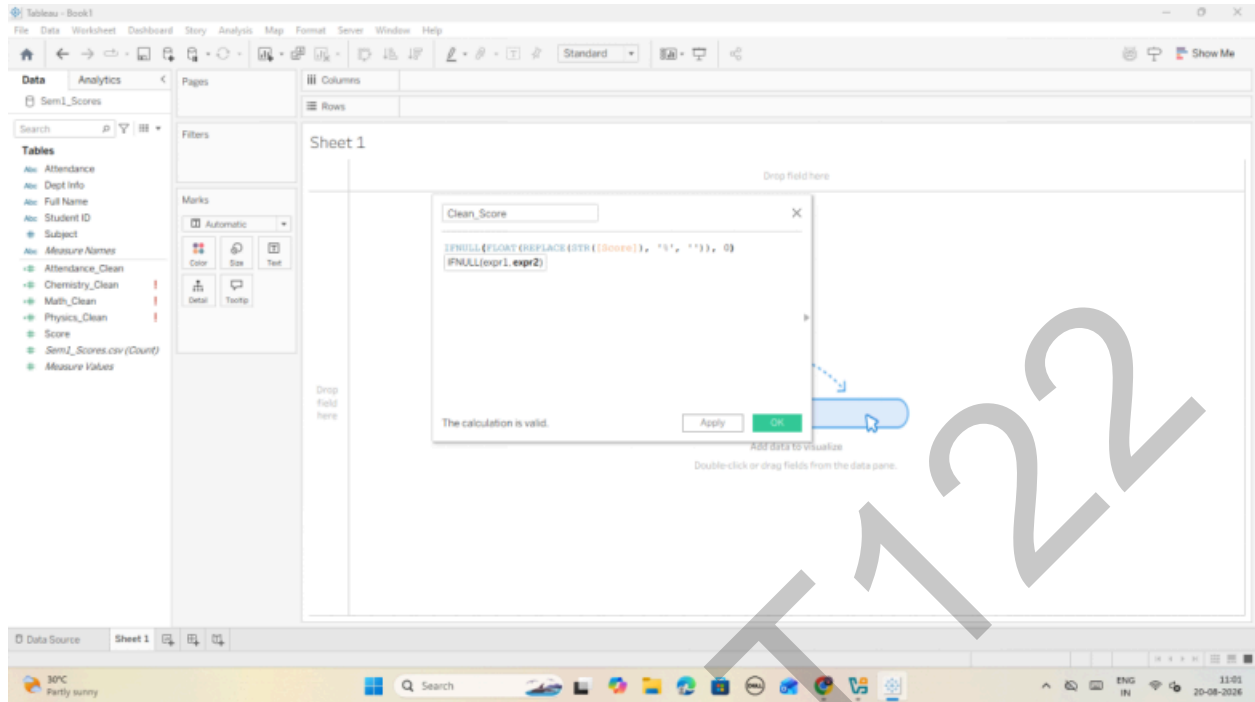
Type	Field Name	Physical Table	Remote Field
Text	Attendance	Sem1_Scores.csv	Attendance
Text	Subject	Pivot	Pivot Field Name
Text	Score	Pivot	Pivot Field Value
Text	Physics_Ck	Calculation	Calculation_D
Text	Attendance_C	Calculation	Calculation_D
Text	Math_Ck	Calculation	Calculation_D

10 fields 40 rows

Student ID	Full Name	Dept Info	Attendance	Subject	Pivot Score	Physics_Clean	Attendance_Clean	Math_Clean	Chems
S001	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	null	85.0000	null	
S001	Aarav Sharma	CSE-01	85 %	Math	86.0000	null	86.0000	null	
S001	Aarav Sharma	CSE-01	85 %	Physics	0.9000	null	85.0000	null	
S001	Aarav Sharma	CSE-01	85 %	Programming	95.0000	null	85.0000	null	
S002	priya VERMA	ECE-02	90 %	Chemistry	80.0000	null	90.0000	null	
S002	priya VERMA	ECE-02	90 %	Math	null	null	90.0000	null	
S002	priya VERMA	ECE-02	90 %	Physics	0.7500	null	90.0000	null	
S002	priya VERMA	ECE-02	90 %	Programming	85.0000	null	90.0000	null	
S003	Rohan Gupta	CSE-01	78 %	Chemistry	78.0000	null	78.0000	null	
S003	Rohan Gupta	CSE-01	78 %	Math	78.0000	null	78.0000	null	

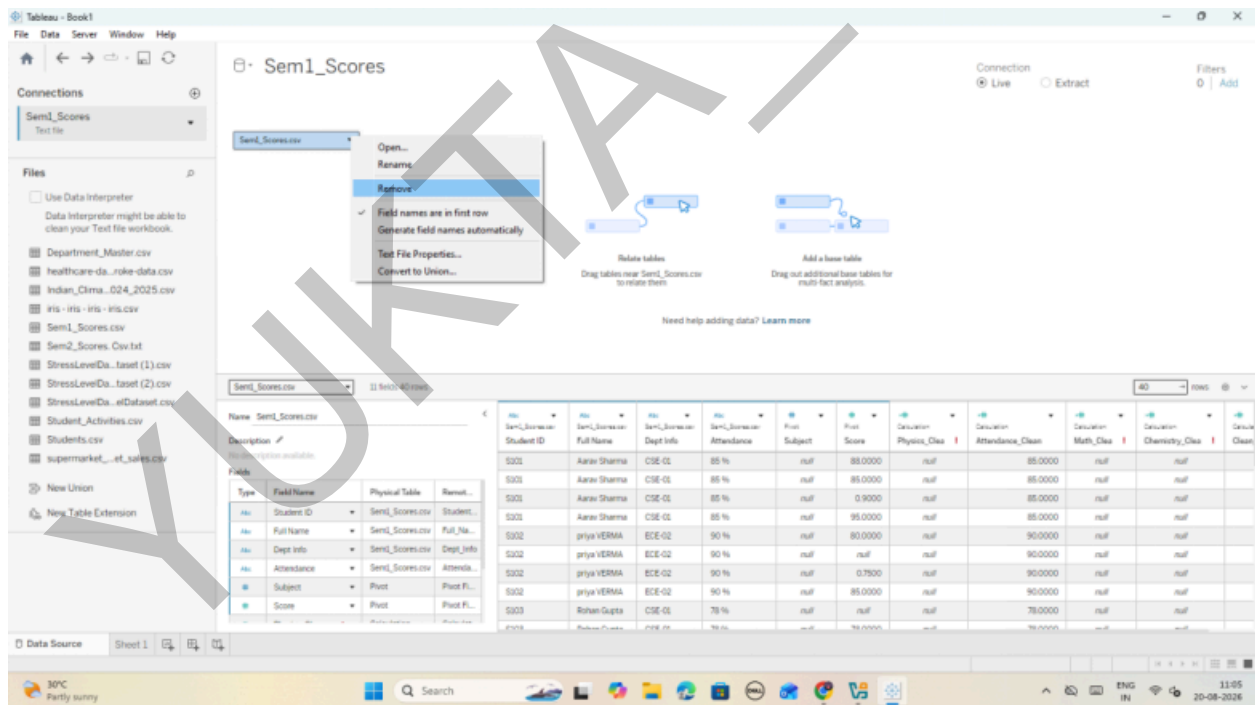
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c) Perform union of multiple files

d) Execute cross-database joins



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The first screenshot shows the 'Union' dialog box in Tableau. The 'Specific (manual)' tab is selected, and two connections, 'Sem1_Scores.csv' and 'Sem2_Scores.csv', are listed. The 'Tables in union: 2' is indicated at the bottom. The 'Apply' button is highlighted.

The second screenshot shows the Tableau interface with the 'Sem1_Scores' data source selected. The 'Fields' pane on the left lists various fields. The main view displays a table with the following data:

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance	PHYSICS_CLEAN
S001	Aarav Sharma	CSE-01	85	0.900000	88	95	85%	0.90000
S002	priya VERMA	ECE-02	null	0.750000	80	85	90%	0.75000
S003	Rohan Gupta	CSE-01	78	0.820000	null	88	78%	0.82000
S004	Sneha Patel	IT-03	92	0.880000	90	96	95%	0.88000
S005	aditya Rao	MECH-04	45	null	50	60	65%	0.00000
S006	Ananya Singh	CSE-01	88	0.980000	85	92	88%	0.98000
S007	Rahul Nair	ECE-02	null	0.700000	75	80	72%	0.70000
S008	Kavya Reddy	IT-03	95	0.940000	92	98	98%	0.94000

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Tableau - T100_PRACT09

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Attendance_students.csv
Categories.csv
Customers.csv
Data Set - In_id_2014.csv
Department_Master.csv
house_price_dataset.csv
housing-housing.csv
Indian_Oli_4_2025.csv
List of Orders.csv
Netflix.csv
Order_Details.csv
Order_Details_chical_3.csv
Orders(1).csv
Retail_sales_all_sales.csv
Retail_sales.csv
Sample - Su - Orders.csv
Sem1_Scores.csv
New Union
New Table Extension

Sem1_Scores.csv

Relate tables
Drag tables near Sem1_Scores.csv to relate them

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? Learn more

Sem1_Scores.csv 12 fields 10 rows

Name: Sem1_Scores.csv
Description: No description available.
Fields: Type Field Name
Dept Info
Math
Physics
Chemistry
Programming

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance	PHYSICS_CLEAN
S001	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %	0.900000
S002	priya VERMA	ECE-02	null	0.750000	80	85	90 %	0.750000
S003	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %	0.820000
S004	Sneha Patel	IT-03	92	0.880000	90	96	95 %	0.880000
S005	aditya Rao	MEDH-04	45	null	50	80	65 %	0.000000
S006	Ananya Singh	CSE-01	88	0.910000	85	92	88 %	0.910000
S007	Rahul Nair	ECE-02	null	0.700000	75	80	72 %	0.700000
S008	Kavya Reddy	IT-03	95	0.940000	92	98	98 %	0.940000

29°C
Partly sunny

Search

ENG IN 10:27 20-08-2026

Tableau - T100_PRACT09

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Sem1_Scores.csv
Sem2_Scores.csv
Sem2_Scores.csv.txt
Sem2_Scores.csv
startup_funding.csv
store_sales_data(2).csv
StressLevel_set(1).csv
StressLevel_Dataset.csv
Student_Activities.csv
Student_Performance.csv
Students.csv
supermarket_sales(2).csv
supermarket_sales.csv
table-1785_716728.csv
table-1785_987032.csv
Titanic-Dataset.csv
Titanic.csv
weatherHistory.csv
New Union
New Table Extension

Sem1_Scores.csv

Relate tables
Drag tables near Sem1_Scores.csv to relate them

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? Learn more

Sem1_Scores.csv 12 fields 40 rows

Name: Sem1_Scores.csv
Description: No description available.
Fields: Type Field Name
Attendance
Pivot Field Names
Pivot Field Values
PHYSICS_CLEAN
Attendance_clean

Student ID	Full Name	Dept Info	Attendance	Pivot Field Names	Pivot Field Values	PHYSICS_CLEAN	Attendance_clean
S001	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	null	85.00
S002	Aarav Sharma	CSE-01	85 %	Math	85.0000	null	85.00
S001	Aarav Sharma	CSE-01	85 %	Physics	0.9000	null	85.00
S001	Aarav Sharma	CSE-01	85 %	Programming	95.0000	null	85.00
S002	priya VERMA	ECE-02	90 %	Chemistry	80.0000	null	90.00
S002	priya VERMA	ECE-02	90 %	Math	null	null	90.00
S002	priya VERMA	ECE-02	90 %	Physics	0.7500	null	90.00
S002	priya VERMA	ECE-02	90 %	Programming	85.0000	null	90.00

30°C
Partly sunny

Search

ENG IN 10:53 20-08-2026

SHETH L.U.J. AND SIR M.V. COLLEGE

Tableau - T100_PRACT09

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Sem1_Scores.csv
Sem2_Scores.Csv.txt
Sem2_Scores.csv
startup_funding.csv
store_sales_data (2).csv
StressLevel...set (1).csv
StressLevel_Dataset.csv
Student_Activities.csv
Student_Performance.csv
Students.csv
supermarket_sales (2).csv
supermarket_sales.csv
table-1785_716728.csv
table-1785_987032.csv
Titanic-Dataset.csv
Titanic.csv
weatherHistory.csv

New Union
New Table Extension

Sem1_Scores.csv

30 fields 40 rows

Need help adding data? Learn more

Student ID	Full Name	Dept Info	Attendance	Subject	Score	PHYSICS_CLEAN	Attendance_clean
S001	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	null	85.00
S002	Aarav Sharma	CSE-01	85 %	Math	85.0000	null	85.00
S003	Aarav Sharma	CSE-01	85 %	Physics	0.9000	null	85.00
S004	Aarav Sharma	CSE-01	85 %	Programming	95.0000	null	85.00
S002	priya VERMA	ECE-02	90 %	Chemistry	80.0000	null	90.00
S002	priya VERMA	ECE-02	90 %	Math	null	null	90.00
S002	priya VERMA	ECE-02	90 %	Physics	0.7500	null	90.00
S002	priya VERMA	ECE-02	90 %	Programming	85.0000	null	90.00

Tableau - T100_PRACT09

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Sem1_Scores.csv
Sem2_Scores.Csv.txt
Sem2_Scores.csv
startup_funding.csv
store_sales_data (2).csv
StressLevel...set (1).csv
StressLevel_Dataset.csv
Student_Activities.csv
Student_Performance.csv
Students.csv
supermarket_sales (2).csv
supermarket_sales.csv
table-1785_716728.csv
table-1785_987032.csv
Titanic-Dataset.csv
Titanic.csv
weatherHistory.csv

New Union
New Table Extension

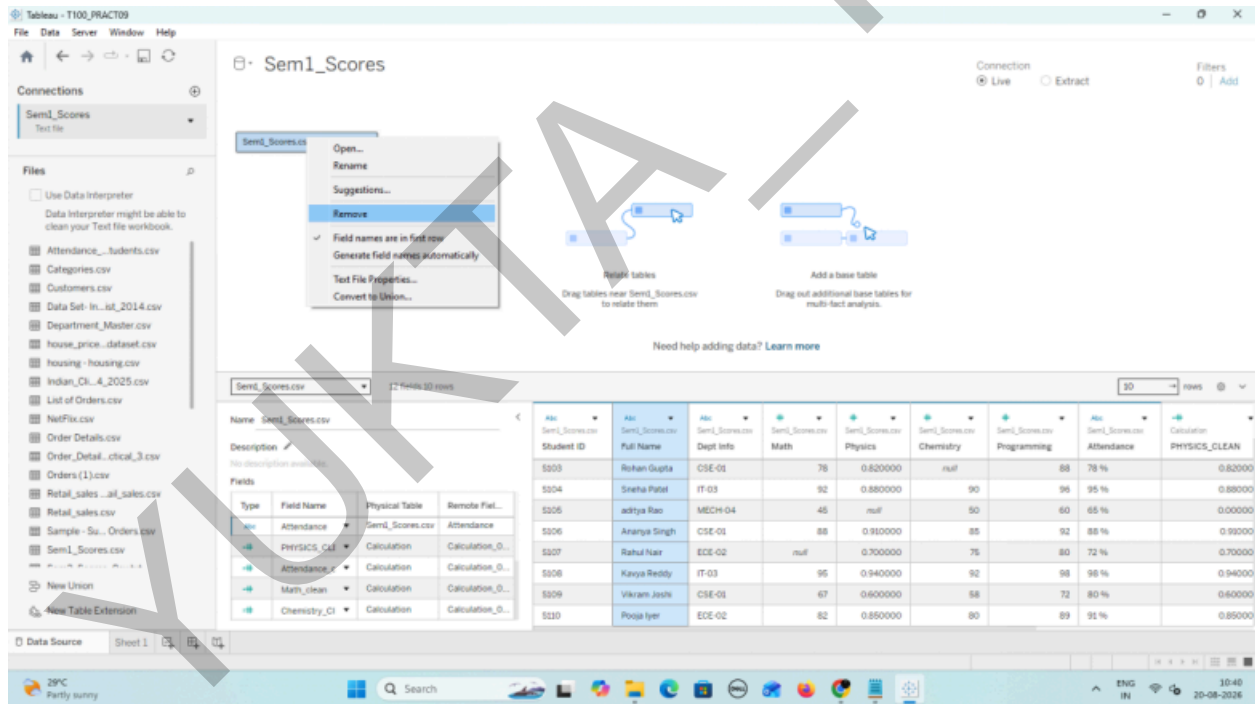
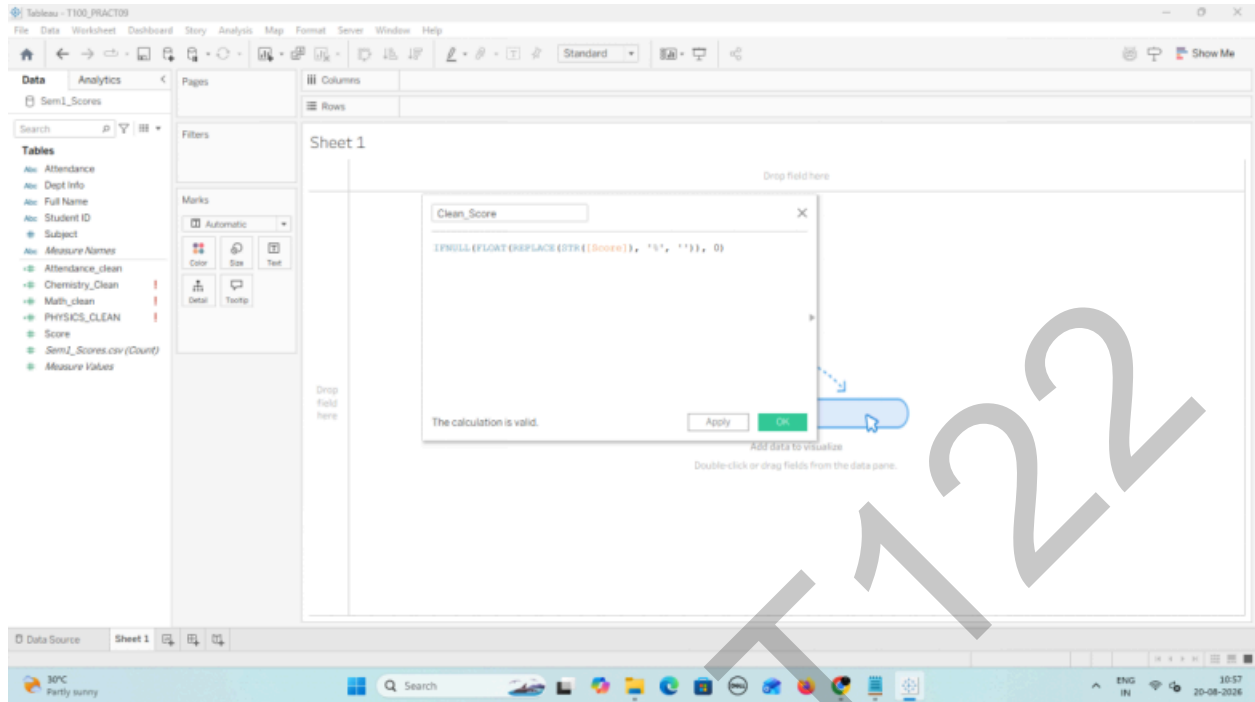
Sem1_Scores.csv

30 fields 40 rows

Need help adding data? Learn more

Student ID	Full Name	Dept Info	Attendance	Subject	Score	PHYSICS_CLEAN	Attendance_clean
S001	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	null	85.00
S002	Aarav Sharma	CSE-01	85 %	Math	85.0000	null	85.00
S003	Aarav Sharma	CSE-01	85 %	Physics	0.9000	null	85.00
S004	Aarav Sharma	CSE-01	85 %	Programming	95.0000	null	85.00
S002	priya VERMA	ECE-02	90 %	Chemistry	80.0000	null	90.00
S002	priya VERMA	ECE-02	90 %	Math	null	null	90.00
S002	priya VERMA	ECE-02	90 %	Physics	0.7500	null	90.00
S002	priya VERMA	ECE-02	90 %	Programming	85.0000	null	90.00

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The top screenshot shows the 'Union' dialog box in Tableau. The 'Specific (manual)' tab is selected, and the 'Connection: Sem1_Scores' is shown. Two files, 'Sem1_Scores.csv' and 'Sem2_Scores.csv', are listed in the 'Tables in union: 2' section. The 'Apply' button is highlighted.

The bottom screenshot shows the Tableau interface with the 'Union' operation applied. The 'Fields' pane on the left lists the fields from both tables: 'Student_ID', 'Full_Name', 'Dept_Info', 'Math', and 'Physics'. The 'Physical Table' column shows the source table for each field. The 'Remote Fl.' column shows the field name in the remote table. The 'Fields' pane also includes a 'Table Name' field.

Type	Field Name	Physical Table	Remote Fl.
Text	Student_ID	Union	Student_ID
Text	Full_Name	Union	Full_Name
Text	Dept_Info	Union	Dept_Info
Text	Math	Union	Math
Text	Physics	Union	Physics

The screenshot displays the Tableau Desktop interface. On the left, the 'Data' pane shows a list of tables including 'Sem1_Scores'. The 'Marks' card is set to 'Automatic'. The main view shows a calculated field named 'Semester' with the following logic:

```

IF CONTAINS([Table Name], "Sem1")
THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2")
THEN "Semester 2" ELSE "Other"
END

```

The calculation is valid. Below the calculated field, there is a table titled 'Sem1_Scores' with the following data:

Ali	Union	Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendance	Table Name
S201	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %	Sem1_Scores.csv		
S202	priya VERMA	ECE-02	null	0.750000	80	85	90 %	Sem1_Scores.csv		
S203	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %	Sem1_Scores.csv		
S204	Sneha Patel	IT-03	92	0.880000	90	96	95 %	Sem1_Scores.csv		
S205	aditya Rao	MEDH-04	45	null	50	60	65 %	Sem1_Scores.csv		
S206	Ananya Singh	CSE-01	88	0.960000	85	92	88 %	Sem1_Scores.csv		
S207	Rahul haar	ECE-02	null	0.700000	75	80	72 %	Sem1_Scores.csv		
S208	Kavya Reddy	IT-03	95	0.940000	92	98	98 %	Sem1_Scores.csv		

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The top screenshot shows the Tableau Desktop interface with the 'Sem1_Scores' data source selected. The 'Connections' pane on the left lists various data sources, including 'Sem1_Scores.csv'. The 'Files' pane on the right shows a list of files, including 'Sem1_Scores.csv'. The 'Data Source' pane on the right shows the 'Sem1_Scores' data source with a 'Union' relationship between 'Student_Activities.csv' and 'Student_Activities.csv'. The 'Columns' shelf is empty, and the 'Rows' shelf is empty. The 'Marks' shelf is empty. The 'Data Source' pane shows a table with columns: Student ID (Student Activities), Club Membership, Projects Completed, and Certifications. The table contains 8 rows of data.

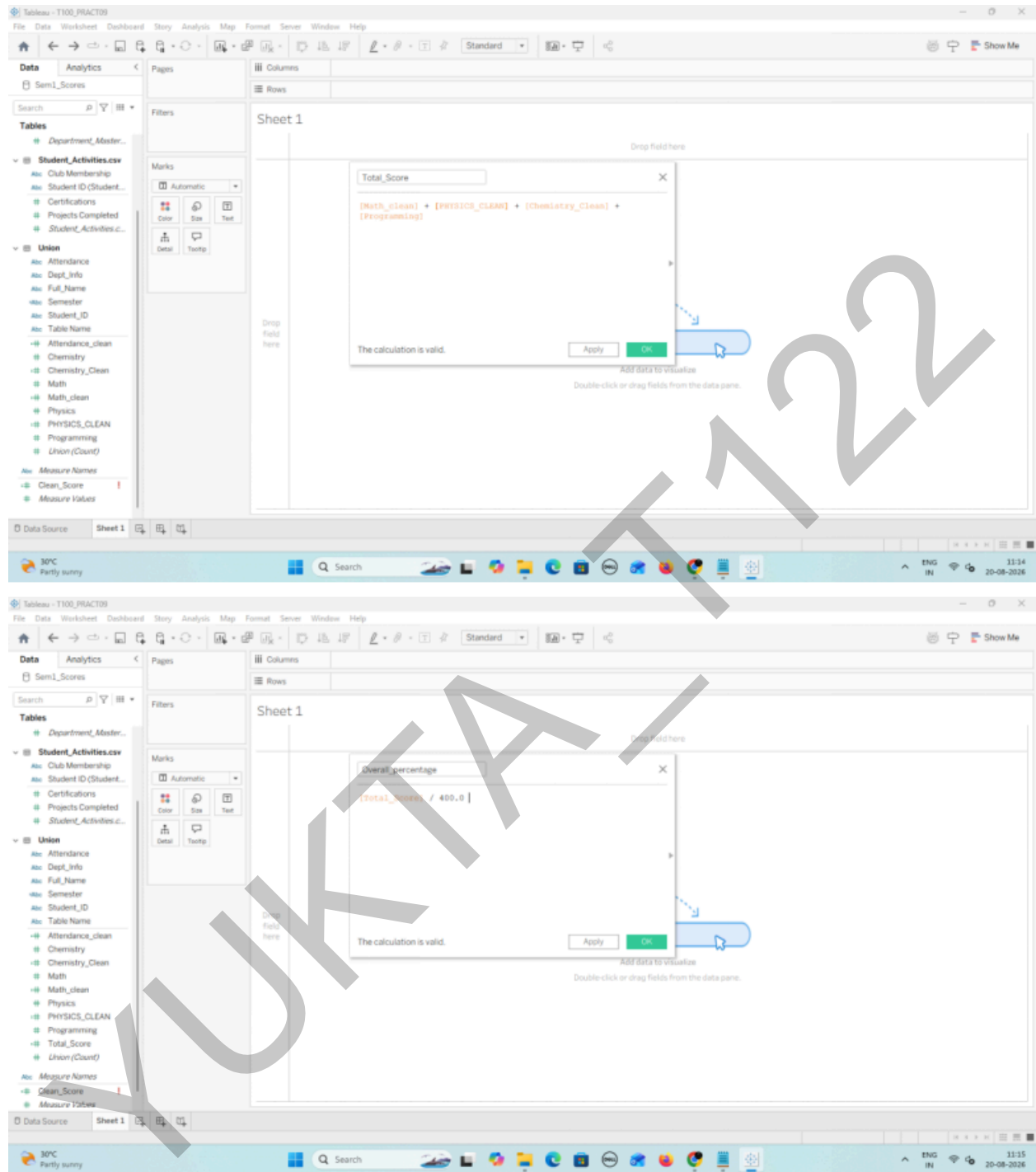
Student ID (Student Activities)	Club Membership	Projects Completed	Certifications
S001	Coding Club	3	2
S002	Robotics Club	2	1
S003	Coding Club	2	1
S004	Debate Society	4	3
S005	Sports Club	1	0
S006	Coding Club	3	2
S007	Robotics Club	2	1
S008	Debate Society	4	3

The bottom screenshot shows the Tableau Desktop interface with a blank worksheet. The 'Data Source' pane on the left shows the 'Sem1_Scores' data source. The 'Columns' shelf is empty, and the 'Rows' shelf is empty. The 'Marks' shelf is empty. The 'Data Source' pane shows a table with columns: Student ID (Student Activities), Club Membership, Projects Completed, and Certifications. The table contains 8 rows of data.

Student ID (Student Activities)	Club Membership	Projects Completed	Certifications
S001	Coding Club	3	2
S002	Robotics Club	2	1
S003	Coding Club	2	1
S004	Debate Society	4	3
S005	Sports Club	1	0
S006	Coding Club	3	2
S007	Robotics Club	2	1
S008	Debate Society	4	3

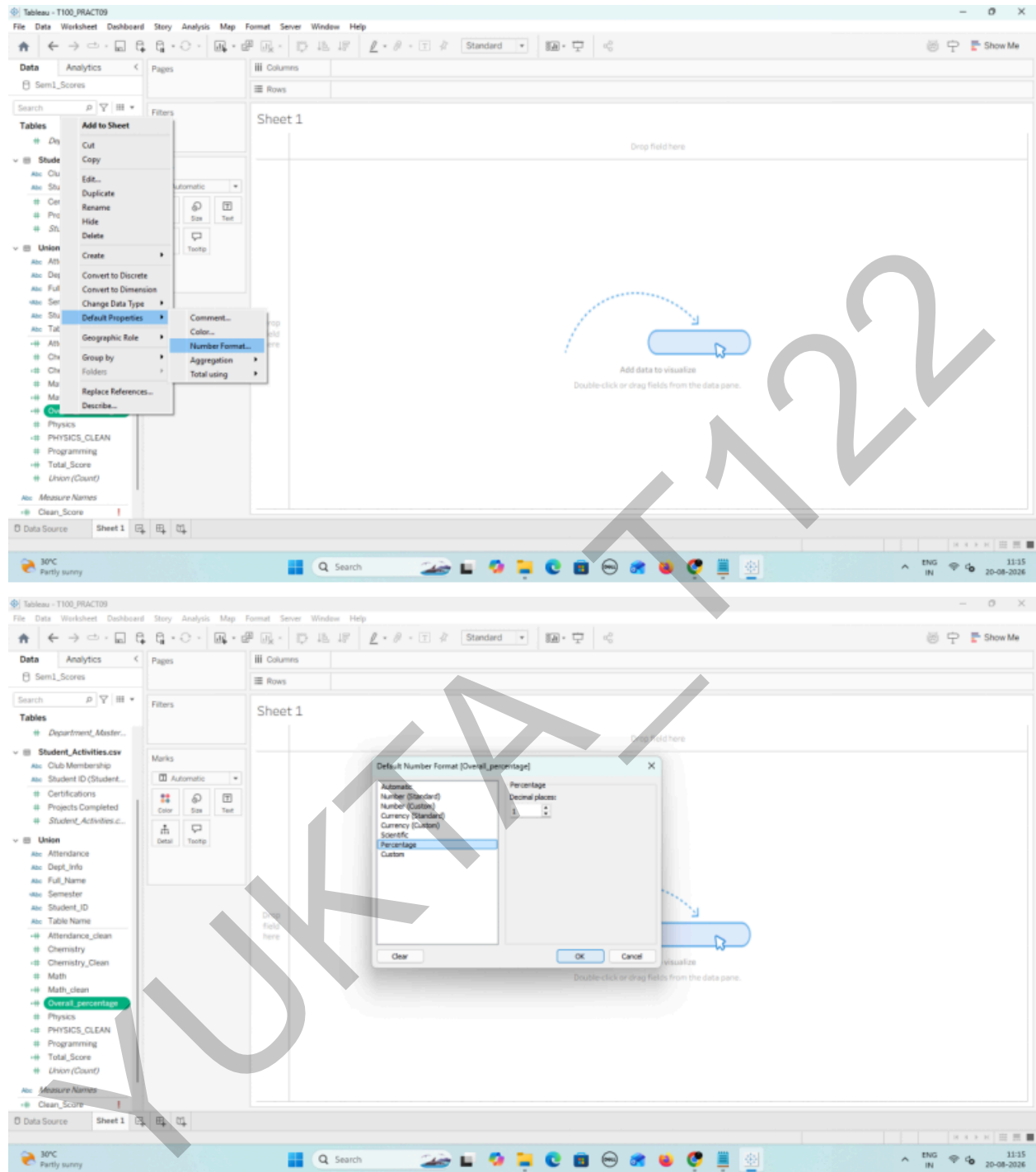
- e) Restructure data for better visualization
- f) Handle different levels of detail in datasets

SHETH L.U.J. AND SIR M.V. COLLEGE



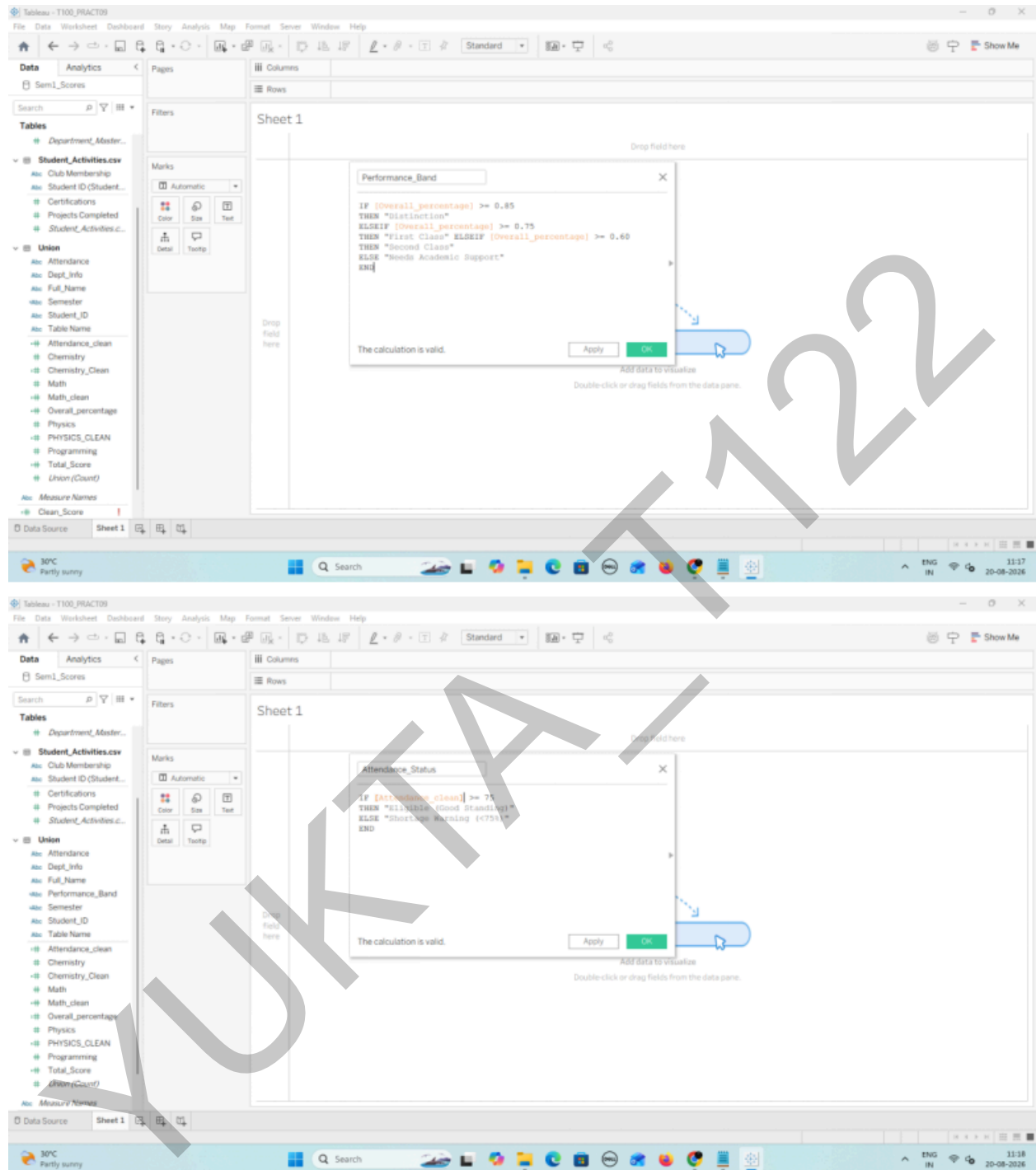
YUKTA SONAWANE T122
PRACTICAL NO : 9

SHETH L.U.J. AND SIR M.V. COLLEGE



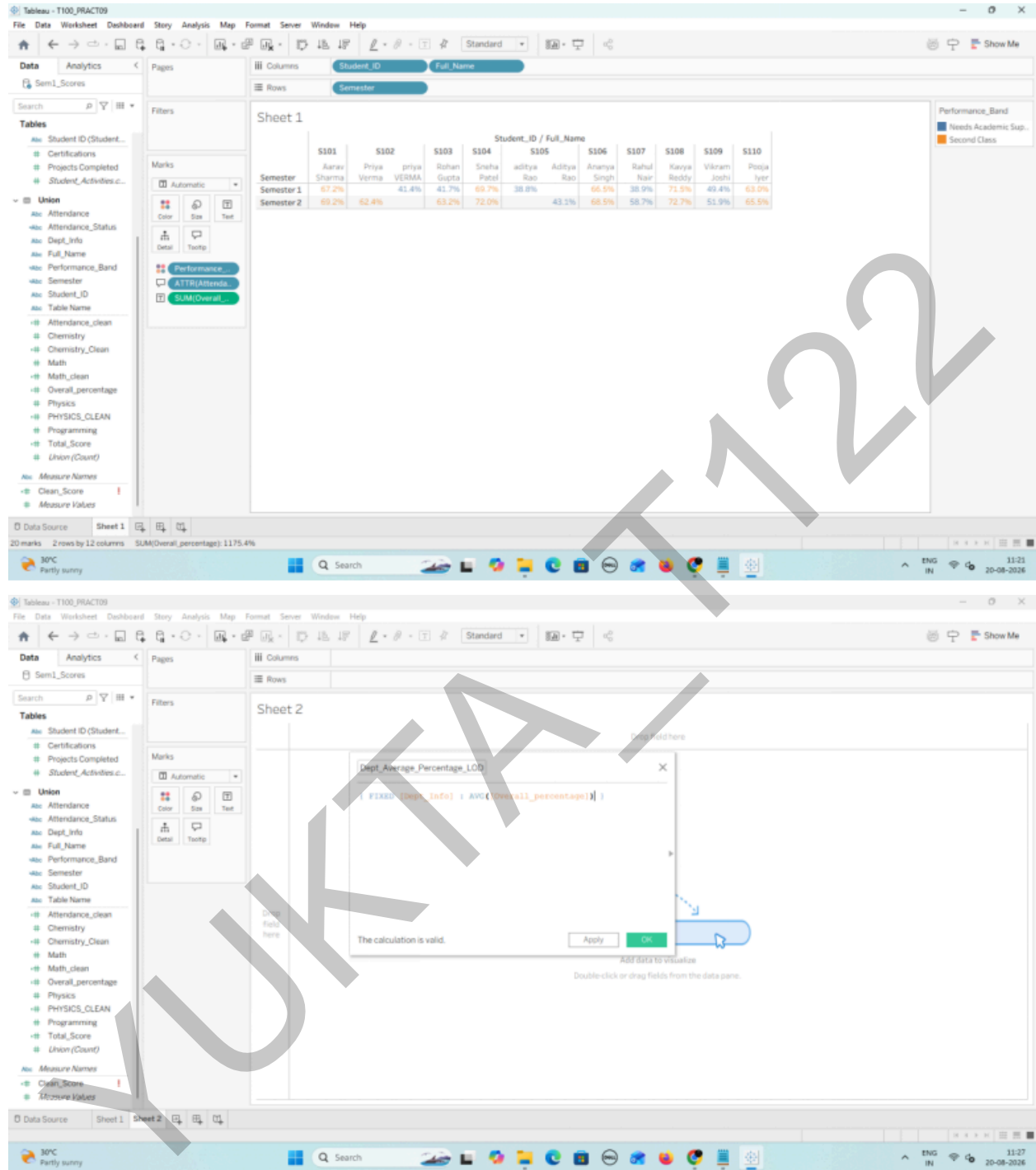
YUKTA SONAWANE T122
PRACTICAL NO : 9

SHETH L.U.J. AND SIR M.V. COLLEGE

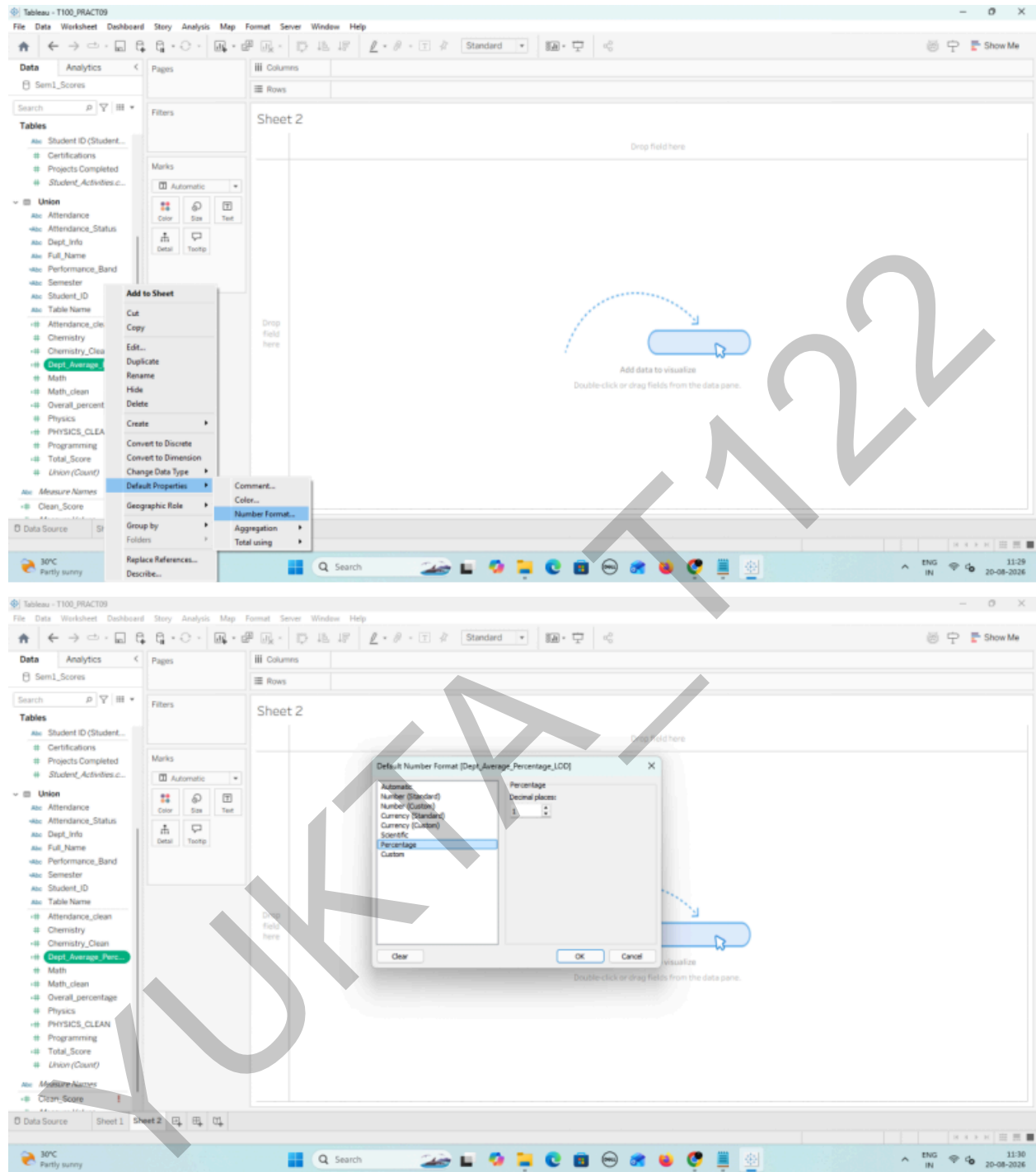


YUKTA SONAWANE T122
PRACTICAL NO : 9

SHETH L.U.J. AND SIR M.V. COLLEGE

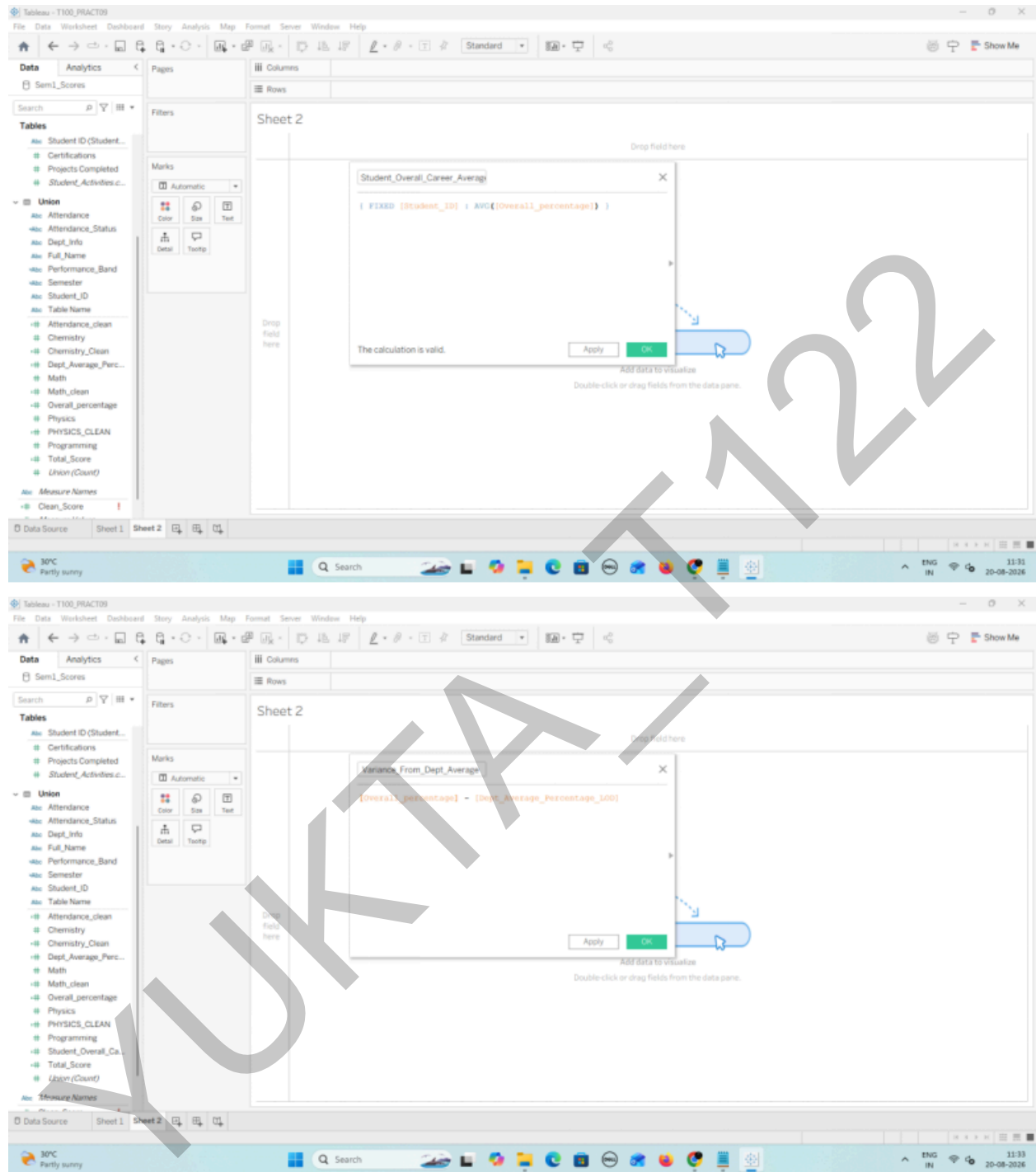


SHETH L.U.J. AND SIR M.V. COLLEGE



YUKTA SONAWANE T122
PRACTICAL NO : 9

SHETH L.U.J. AND SIR M.V. COLLEGE



YUKTA SONAWANE T122
PRACTICAL NO : 9

SHETH L.U.J. AND SIR M.V. COLLEGE

The first screenshot shows the Tableau interface with a calculated field dialog box open. The field is named 'Benchmark_status' and contains the following logic:

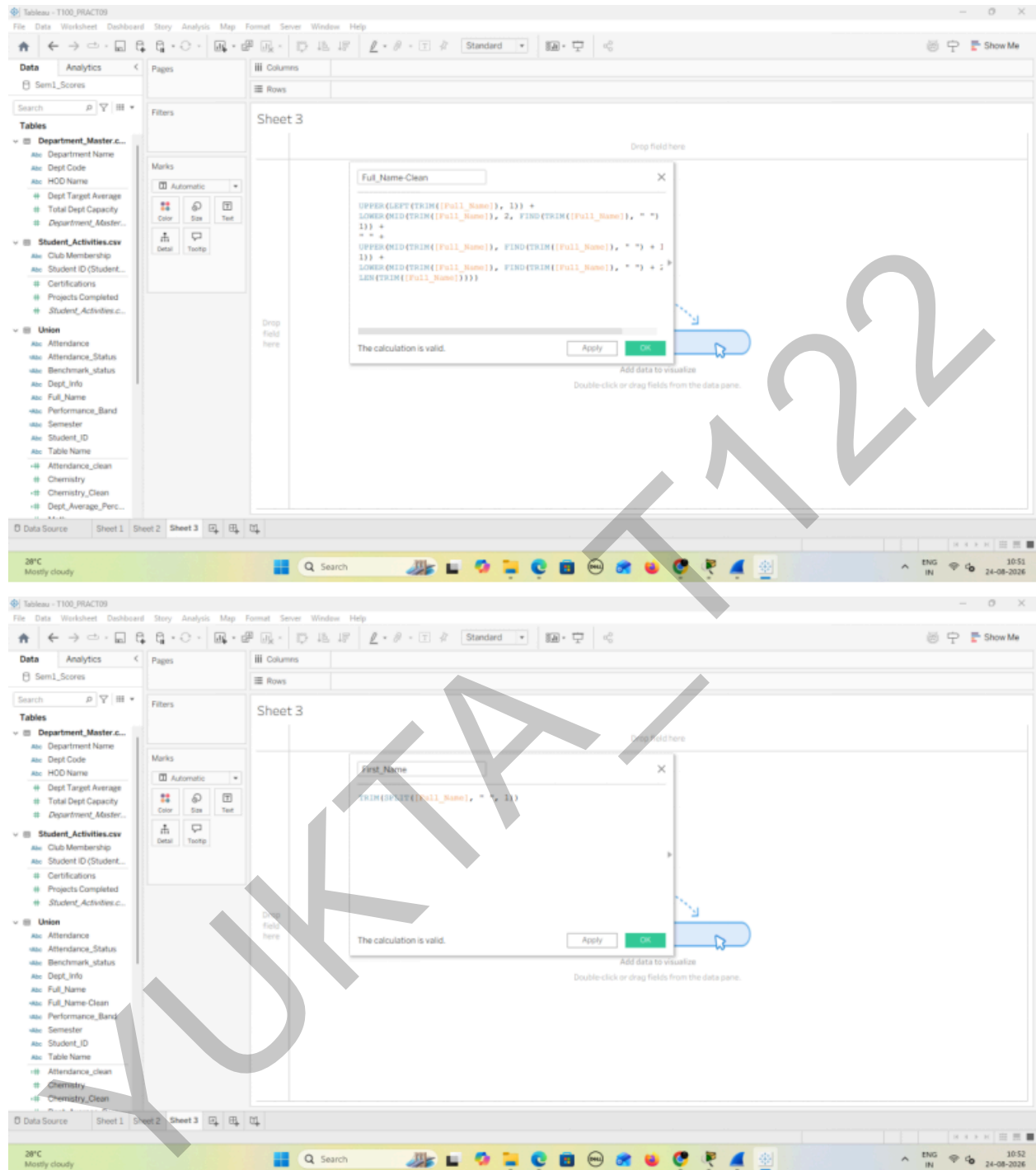
```
IF [Variance_From_Dept_Average] >= 0  
THEN "Above Department Benchmark"  
ELSE "Below Department Benchmark"  
END
```

The second screenshot shows the final visualization in a table format. The columns are 'Dept_Info', 'Student_ID', 'Full_Name', and 'Semester'. The 'Semester' column is further broken down into 'Semester r1' and 'Semester r2'. The 'Benchmark_status' is color-coded: blue for 'Above Department B.' and orange for 'Below Department B.'.

Dept_Info	Student_ID	Full_Name	Semester		Benchmark_status
			Semester r1	Semester r2	
CSE-01	S101	Anay Sharma	67.2%	69.2%	Above Department B.
	S103	Rishabh Gupta	41.7%	63.2%	Below Department B.
	S106	Ananya Singh	66.5%	68.9%	Above Department B.
	S109	Vikram Joshi	59.4%	51.9%	Below Department B.
ECE-02	S102	Priya Verma		62.4%	
	S107	Priya VERMA	61.4%		
IT-03	S120	Rahul Nair	59.8%	58.7%	Below Department B.
	S124	Pooja Iyer	69.7%	72.0%	Above Department B.
	S128	Sneha Patel	69.7%	72.7%	Above Department B.
MECH-04	S105	Kavya Reddy	71.5%	72.7%	Above Department B.
		Aditya Rao	58.8%	49.1%	Below Department B.

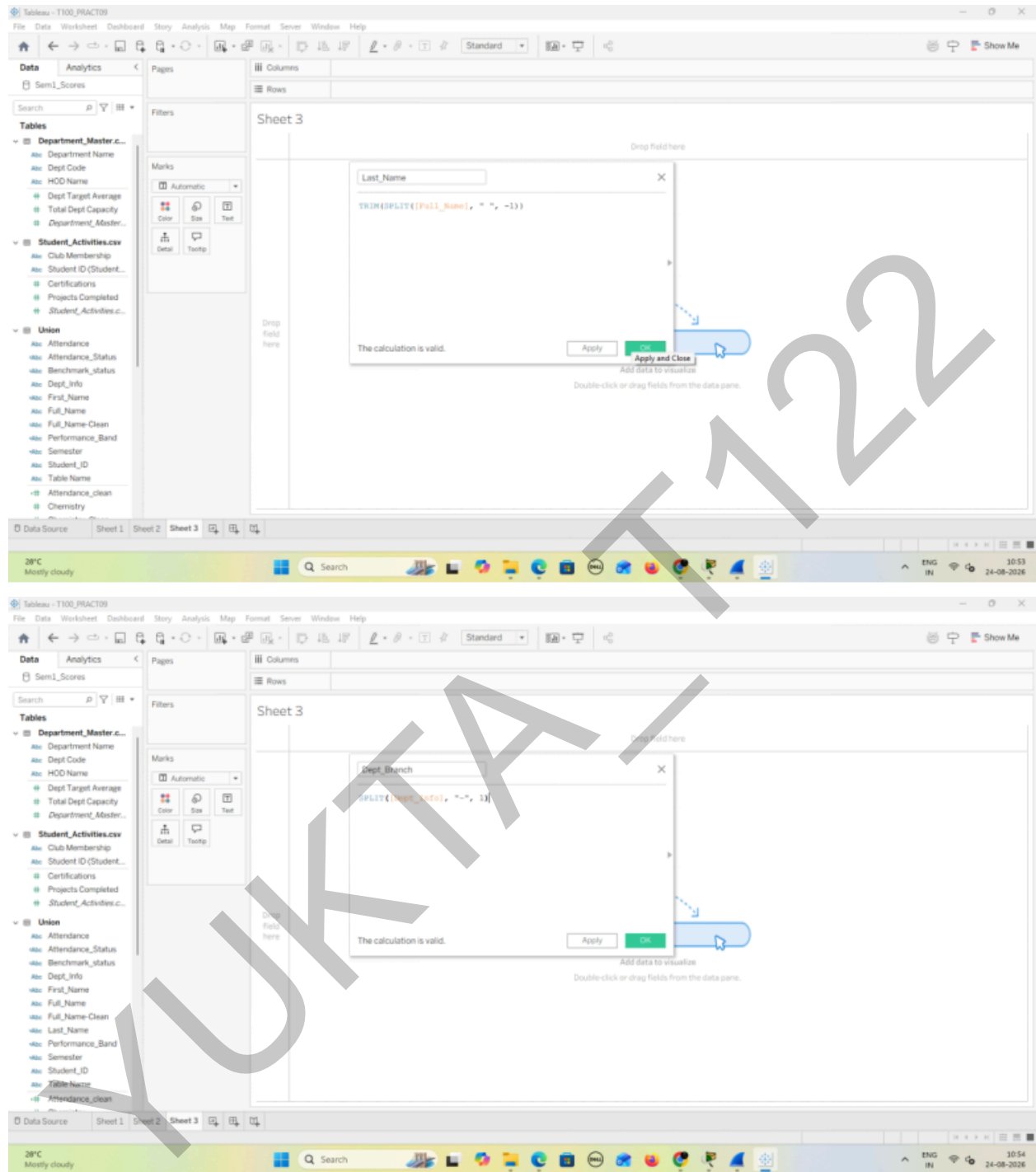
g) Apply advanced data cleaning techniques

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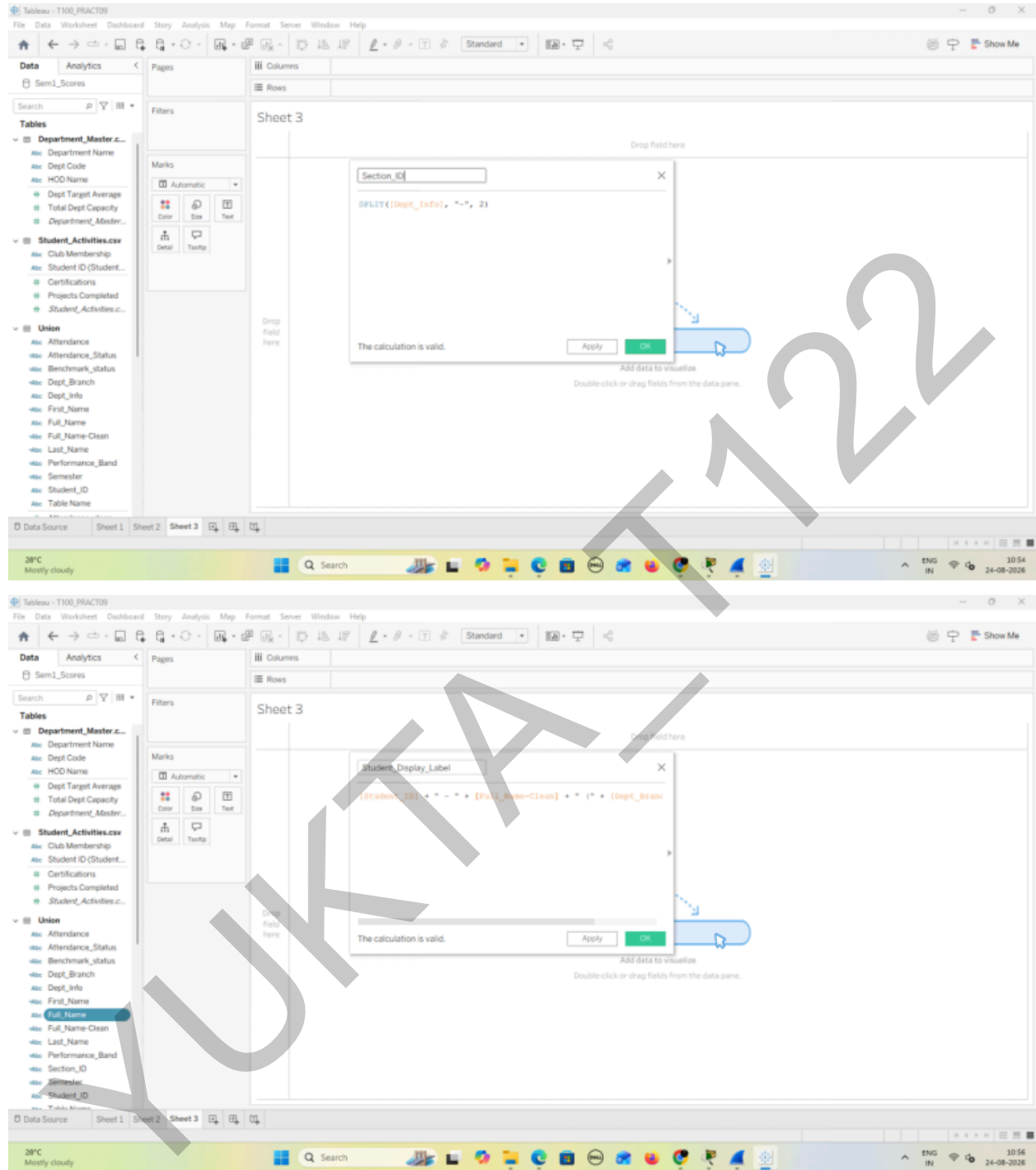
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